DSGE Model Estimation via VAR IRF Matching: The Short and Long of It *

Lorenzo Del Col[ξ]

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Abstract

Following De Graeve and Westermark (2025) and Baumeister (2025), I increase the lag length of the VAR in Altig et al. (2005) to re-estimate their DSGE. Moving from $p = 4$ to $p = 12$ lags, I find that both the VAR and DSGE impulse responses, and the implied structural parameters, change substantially. The Calvo price parameters that distinguish the observationally equivalent firm-specific and homogeneous capital versions of the model converge to one another, from 0.34 against 0.82 at $p = 4$ to 0.89 against 0.90 at $p = 12$. The implied duration of price contracts in the firm-specific model rises from 1.5 to 9.0 quarters, equal to that of the homogeneous capital model, re-opening the micro–macro discrepancy in price stickiness that firm-specific capital was introduced to resolve.

Keywords: Truncation Bias, Long-Lag VARs, DSGE Estimation, IRF Minimum Distance , Nominal Rigidities
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[ξ]University of Padova. Contact: lorenzo.delcol@studenti.unipd.it

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1 Introduction

1.1 Motivation

An important goal of monetary business cycle research is to explain the empirical evidence on the impact of economic shocks on macroeconomic variables. Because the systematic component of policy is endogenous, identifying the causal effect of monetary policy requires isolating the non-systematic component, the monetary policy shock, even though it is small and infrequent (Ravenna, 2007; Aruoba and Drechsel, 2024).

A large class of estimated DSGE models is taken to the data through this channel. Impulse Response Function Matching Estimation (IRFME) is a limited-information strategy that chooses structural parameters to minimise the distance between the impulse responses implied by the model and those estimated from an identified VAR (Christiano et al., 2005; Altig et al., 2011; Rossi et al., 2012). The strategy is attractive precisely because it is partial: the researcher takes a stand on the identification of a subset of shocks, and remains agnostic about the remaining dimensions of the data.

That attractiveness rests entirely on one maintained assumption: that the VAR delivers *consistent* estimates of the moments being matched. If the reduced-form VAR adequately captures the data generating process, impulse responses are optimal at all horizons; if the VAR is misspecified, the specification error propagates to the structural statistics and compounds at each horizon (Ramey, 2016; Canova and Fosso, 2025) Ravenna (2007) "The bias introduced by truncation can lead to bias in the identification of the structural shock." Altig et al. (2011) say as much themselves: "the quality of our estimation strategy depends on the ability of identified VARs to generate reliable estimates of the dynamic response of economic variables to shocks".[1]

Since a VAR(∞) cannot be estimated on a finite sample, researchers truncate. The resulting VAR(p) is misspecified, and increasingly so the smaller is p (De Graeve and Westermark, 2025; Barnichon et al., 2026; Ravenna, 2007). Ravenna (2007) "the problem facing economistis is whether the number of lags p to be included is reasonable given the lenght of economics time series over which VARs are estimated. When estimating a VAR consistent with DSGE business cycle models it is standard to assume that including few lags is sufficient to provide a reasonable approximation to the exact infinite order VAR representation! De Graeve and Westermark (2025) show that the conventional objection to longer lags, that the additional parametrisation makes confidence intervals prohibitively wide, is incomplete, because reducing misspecification reduces not only bias but also variance. Their conclusion is addressed explicitly to applied researchers: "it is up to the researcher to show that her structural results are robust to significant increases in the lag length, far beyond the lag lengths typically considered in current practice".

[1] Altig et al. (2011, footnote 10, p. 3).

1.2 Contribution and Results

This paper asks what lag truncation does to *structural* estimates, not merely to impulse responses. I replicate Altig et al. (2011) — henceforth ACEL — and re-estimate their DSGE by IRF matching against VARs of increasing lag order.

The contribution is twofold. First, relative to De Graeve and Westermark (2025), I move the question from the reduced form to the structural parameters. Second, and more specifically, I do so in a setting their design deliberately excludes. De Graeve and Westermark (2025) state that "all our experiments are based on two-shock models and two-variable VARs", and that where a candidate model does not fit that framework — Smets and Wouters — "we consider the model with only monetary policy and preference shocks".[2] Their data generating processes are square and fundamental by construction, so that truncation bias is isolated from singularity and invertibility. ACEL's model is not: it confronts three structural shocks with a ten-variable VAR. [**TBD:** Verificare che nessun altro abbia gia' fatto questo esercizio su un modello stocasticamente singolare. Se confermato, e' il claim centrale.]

My main results are the following. *First*, the impulse responses to a monetary policy shock retain their sign and broad shape across lag orders, but the humps peak substantially later as p increases, and the point estimates remain inside the VAR(4) confidence bands. *Second*, the responses to the two technology shocks — identified with long-run rather than short-run restrictions — do not: several fall outside the baseline confidence set. This asymmetry is what Ravenna (2007) predicts, since long-run restrictions require the sum of the entire lag polynomial and truncation error accumulates in exactly that object. *Third*, the structural estimates move. The Phillips curve slope falls from 0.0402 to 0.0129 and becomes precisely estimated only at $p = 12$; the cost of varying capital utilisation collapses from 1.99 to 0.004; the investment adjustment cost roughly doubles. *Fourth*, and as a consequence, the implied Calvo price parameters of the firm-specific and homogeneous capital models converge, from 0.34 against 0.82 to 0.89 against 0.90. The implied duration of price contracts in the firm-specific model rises from 1.5 to 9.0 quarters.

The last result overturns ACEL's central conclusion. Their claim is that only the firm-specific capital model reconciles macro evidence on inflation inertia with micro evidence on the frequency of price adjustment. Under a richer lag structure that reconciliation disappears. The channel is transparent and internal to their own argument: ACEL note that "to the extent that capital utilization rates can easily be varied, the assumption of firm-specific capital loses its ability to rationalize low values of ξ_p".[3] Long-lag VARs push the estimate of σ_a precisely into that region.

[2] De Graeve and Westermark (2025, p. 10). They also note that "invertibility is never a problem; all the models we consider are fundamental".

[3] Altig et al. (2011, p. 227).

1.3 Related Literature

Long-lag VARs.
Truncation bias and the DSGE–VAR mapping.
Impulse response matching and DSGE estimation.

1.4 Outline

Section

2 The Theoretical Model

2.1 Inflation inertia and the micro–macro conflict

Altig et al. (2011) start from a tension in the New Keynesian literature. To account for inertia in inflation, standard models require firms to re-optimise prices infrequently — every six quarters, and in versions without indexation to past inflation even every two years. This sits uneasily with the microeconometric evidence, which finds price changes considerably more frequent. [**TBD:** Citare Bils and Klenow (2004), Golosov and Lucas (2007), Klenow and Kryvtsov (2008), che sono i riferimenti di ACEL. Valutare se aggiungere evidenza micro piu' recente. Da aggiungere a biblio.bib.]

Altig et al. (2005)[4] propose a resolution. In their model inflation is inertial even though firms re-optimise on average every 1.5 quarters. Inertia arises not because firms change prices rarely, but because when they do they change them by a small amount: each firm's short-run marginal cost is increasing in its own output, which dampens the optimal price adjustment. The underlying assumption that drives this result is that capital is predetermined within the period and firm-specific, so that factors of production cannot be instantly and costlessly reallocated across firms.

2.2 Homogeneous and firm-specific capital

Altig et al. (2005) conduct the analysis using two versions of the Christiano et al. (2005) model: one in which capital is homogeneous, as in standard formulations, and the other in which it is firm-specific.

Once log-linearised, the only difference between the two sets of equilibrium conditions lies in the equation relating inflation to marginal cost, the New Keynesian Phillisp Curve,

$$\Delta\hat{\pi}_t = \mathbb{E}\big[\beta\Delta\hat{\pi}_{t+1} + \gamma\hat{s}_t \mid \Omega_t\big], \tag{1}$$

[4]The replicaiton code made available from the authors is NBER working paper from 2005. Hence, I replicate and refere to those results.

where $\hat{s}_t$ is economy-wide average real marginal cost in units of the final good, and the slope γ maps into the structural parameters as

$$\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\,\chi. \tag{2}$$

The information set Ω_t contains the current realisation of the technology shocks but not that of the monetary policy innovation. The two models are parametrised in terms of γ rather than ξ_p.

Given the remaining parameters they are then observationally equivalent with respect to aggregate prices and quantities, so that no stand need be taken on which version is being estimated. What distinguishes them is the value of ξ_p given γ implies.

The real rigidity parameter χ measures the strategic complementarity in price setting of Ball and Romer (1990). It takes values in $(0, 1]$, and the closer it lies to zero the stronger the attenuation. That is, a firm that is able to reset its price moves it by less, for a given movement in economy-wide marginal cost. In the homogeneous capital model $\chi = 1$ and the attenuation is absent. In the firm-specific model χ is a non-linear function of the structural parameters.

In particular χ depends on some calibrated parameters, $[a, \beta, \delta, \lambda_f]$ which are respectively, capital component in the Cobb-Douglas production function, time preference parameter, capital depreciation rate, and the average price markup. And two estimated parameters, $[\sigma_a; S'']$, which are the Capacity utilisation cost and Investment adjustment cost.

The Altig et al. (2011) logic suggest that, for a fixed value of ξ_p, the larger σ_a, the lower is γ, in line with the vast set of literature that suggests that the Phillips curve slope is low. But other things equal, a lower ξ_p implies a higher γ. This determine that for a given value of γ , ξ_p is a decreasing function of σ_a.
In this sense, a flat Phillips curve is a problem in the homogenoues capital model. A low value of γ implies that, given that there $\chi = 1$, the Clavo price stickiness parameter ξ_p need to be very high, implying irrealistic implied time duration of prices which are in conflict with micro evidence.
The flexibility of capital utilization is governed from σ_a parameter, and $1/\sigma_a$ is the elasticity of capacity utilization with respect to the rental rate of capital. The quality of the estimation of the firm specific capital model lies on having a high value of capacity utilisation cost of capital σ_a in a way to obtain a low value of price stikyness.
In homogeneous capital model, firms can adjust their level of capital freely, and suggest that should be σ_a close to 0 with the elasticiy being close to infinity. In the stark opposit direction, if the capital is firm specific, firms should have some friction in adjusting its level, hence, σ_a should be strongly greater than 0 implying a rigid elasticity.

This is where the mechanism is fragile, and Altig et al. (2011) say themselves so: "to the extent that capital utilization rates can easily be varied, the assumption of firm-specific capital loses its ability to rationalize low values of ξ_p".[5] The reconciliation of micro and macro evidence rests, in the end, lies on a single estimated parameter.

3 Truncation Bias and the Case for Long-Lag VARs

VARs are subject to at least three sources of bias: small-sample bias, identification bias (Forni et al., 2016), and truncation bias (De Graeve and Westermark, 2025; Ravenna, 2007; Chari et al., 2008; Barnichon et al., 2026). This paper is concerned with the latter, and with the way it contaminates the second.

3.1 A simple illustration

Following De Graeve and Westermark (2025), consider a data generating process in which y_t depends on two exogenous variables,

$$y_t = x_{1t}\beta_1 + x_{2t}\beta_2 + \epsilon_t. \tag{3}$$

Estimating the relationship while omitting x_{2t},

$$y_t = x_{1t}b_1 + e_t, \tag{4}$$

delivers a biased point estimate unless the regressors are orthogonal, $\mathbb{E}(b_1) \neq \beta_1$, and an upward-biased residual variance, $s^2 > \sigma^2$. Both distortions matter here, and the second is the one usually forgotten.

The same logic carries over to a VAR. Let

$$Y_t = B_1 Y_{t-1} + \cdots + B_p Y_{t-p} + u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma_u, \tag{5}$$

with $B(L) = B_1 L + \cdots + B_p L^p$ and $p \ll p^*$, where p^* denotes the true lag order. Truncation biases both the autoregressive coefficients $B(L)$ and the reduced-form covariance matrix Σ_u. Since impulse responses are a function of both,

$$\text{IRF} = f\big(B(L), \Sigma_u\big), \tag{6}$$

they inherit the bias.

[5] Altig et al. (2011, p. 227).

4 Empirical Results

4.1 DSGE structural parameters

Table 1: DSGE Structural Parameters Comparison Across Lag Lengths

Parameter	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Structural Estimated Frictions			
Price markup (λ_f)	1.0100	1.0100	1.0100
	(-)	(-)	(-)
Wage stickiness (ξ_w)	0.7221***	0.5617***	0.4838***
	(0.1506)	(0.1142)	(0.0757)
Phillips curve slope (γ)	0.0402	0.0402	0.0129***
	(0.0246)	(0.0598)	(0.0042)
Capacity util. cost (σ_a)	1.9938	0.1175	0.0042
	(2.6066)	(0.6299)	(0.0230)
Habit formation (b)	0.7066***	0.6935***	0.8596***
	(0.0463)	(0.0570)	(0.0203)
Inv. adj. cost (S'')	3.2807*	3.0372*	6.8431***
	(1.8874)	(1.5506)	(2.3306)
Money demand elast. (ϵ)	0.8077***	0.7531***	0.8346***
	(0.2111)	(0.1473)	(0.1944)
Implied Price Stickiness			
Homogeneous capital ($\xi_{p,hom}$)	0.8213	0.8213	0.8958
	(0.0479)	(0.0574)	(0.0139)
Firm-specific capital ($\xi_{p,fs}$)	0.3372	0.6448	0.8884
	(0.1503)	(0.1285)	(0.0220)

Note: Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The Calvo price parameters $\xi_{p,hom}$ and $\xi_{p,fs}$ are not estimated directly: they are recovered numerically from the estimated structural parameters. Their standard errors are therefore obtained by Monte Carlo simulation with 1,000,000 draws from the joint distribution of the estimated structural parameters, solving for the implied Calvo parameter at each draw.

4.2 Implied duration and the collapse of the firm-specific mechanism

Table 2: Implied Average Time between Re-Optimization

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model			
Calvo parameter ($\xi_{p,hom}$)	0.82	0.82	0.90
Implied duration (quarters)	**5.6**	**5.6**	**9.6**
Firm-Specific Capital Model			
Calvo parameter ($\xi_{p,fs}$)	0.34	0.64	0.89
Implied duration (quarters)	**1.5**	**2.8**	**9.0**

- **Baseline VAR ($p = 4$)**: Homogeneous capital implies price changes every **5.6 quarters**.
- **Firm-specific capital**: Drastically reduces this to **1.5 quarters**, matching microeconomic evidence.

Note: Implied duration is calculated as $D = 1/(1 - \xi_p)$.

The Chi parameter

Table 3: Implied Strategic Complementarity Parameter χ Across Lag Lengths

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model (validity check, $\chi \equiv 1$)			
χ_{hom}	0.9989	1.0010	0.9977
	(0.0000)	(0.0000)	(0.0000)
Firm-Specific Capital Model			
χ_{fs}	0.0307	0.2030	0.8671
	(0.0596)	(0.1667)	(0.1698)

Note: χ is recovered by inverting $\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\chi$ at the estimated γ and implied ξ_p. $\chi_{hom} \equiv 1$ by construction (Proposition 1(i) in Altig et al., 2011); reported here only as a check that the numerical pipeline is consistent. Standard errors in parentheses are computed from the same 1,000,000-draw joint Monte Carlo used for the Calvo parameters' standard errors, applying the formula above draw-by-draw. The 90% Monte Carlo percentile interval for χ_{fs} is [0.0067, 0.0879], [0.0516, 0.5101], [0.3704, 0.9270] for $p = 4, 8, 12$ respectively; see the discussion of non-normality of the Calvo/chi Monte Carlo distributions in the text.

5 Conclusion

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A Data

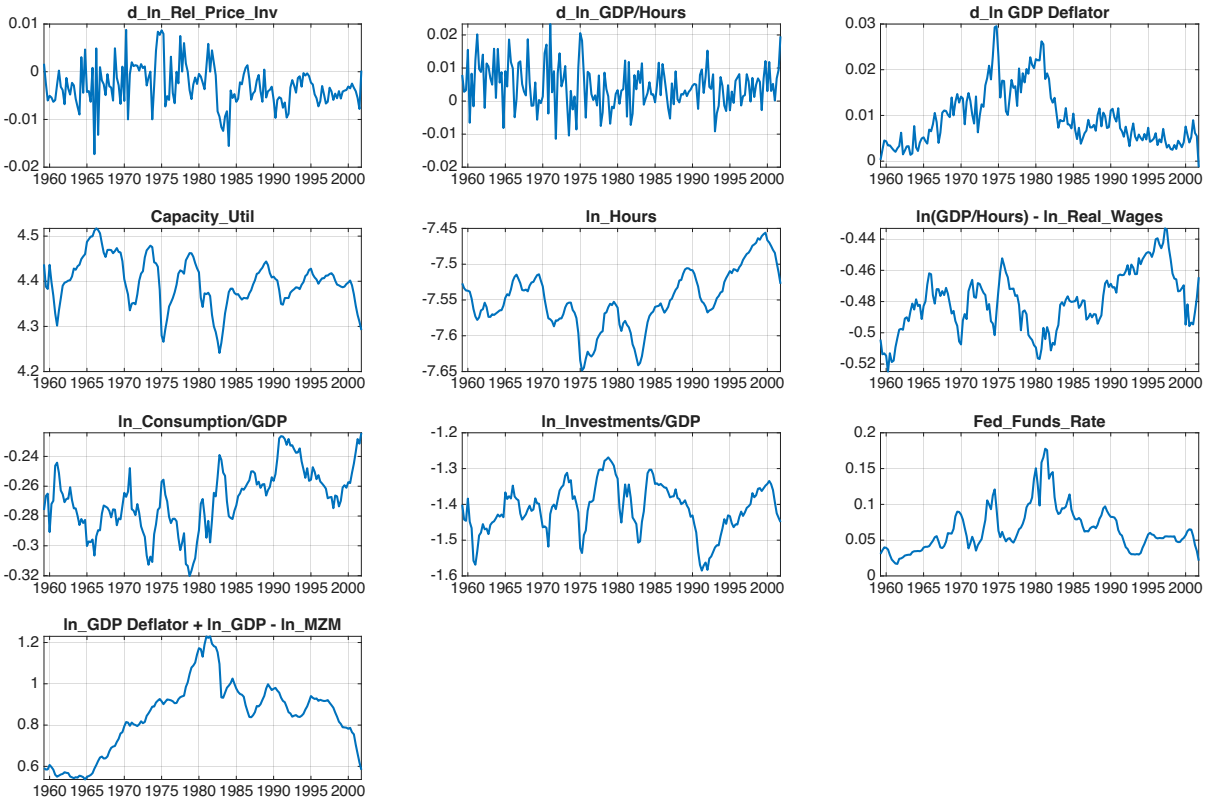


Figure 1: Raw historical series

B The ACEL Model: Full Derivation

C Additional Residual Analysis

D Monte Carlo Inference on the Calvo Parameters

E Robustness

F Replication Files and Code